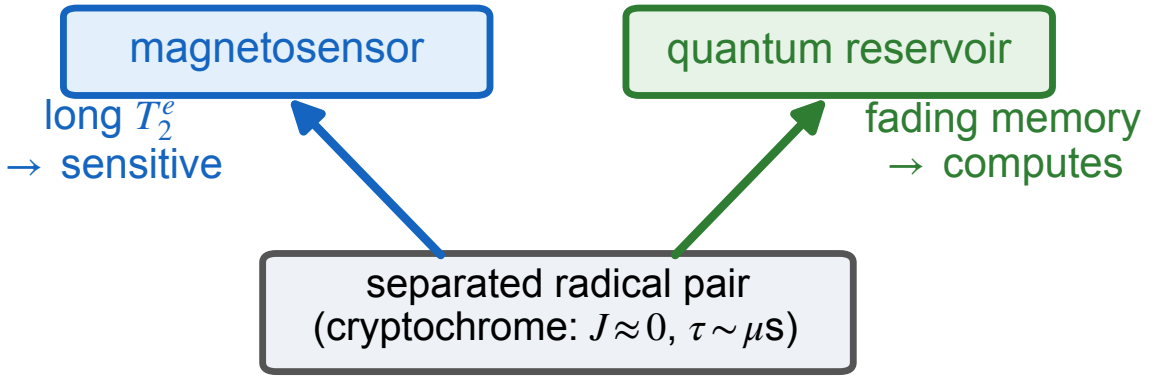
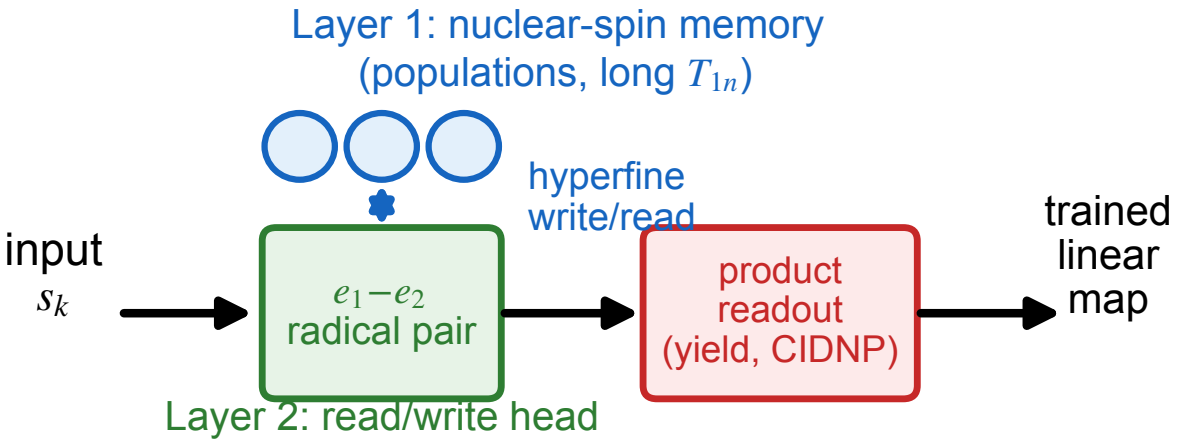
(a) one molecule, two functions



the open cycle — not electronic decoherence — supplies the fading memory

(b) nuclei are the register that must persist



Layer 3: classical readout

$$\text{memory horizon} = (\text{MC} - C_0) \times T_{\text{turnover}}$$

reaches ms–s only if the register is reused *and* survives:

proton-carried (^{14}N is quadrupolar), $\tau_{\text{eff}} \lesssim 2.6$–$4.0$ ns,
catalytic turnover 10–100 ms — not light-driven